

Portable Dual-Gamma Backscatter Landmine Detector

Simulation Validation, Sensor Architecture, and Prototype Readiness Report

Detector Development & Simulation Group

Phase II Milestone & Technical Progress Deck

August 23, 2026

Physical Working Principle & Detection Concept

The Operational Challenge:

- Conventional metal detectors fail against minimum-metal / plastic AP mines.
- Ground Penetrating Radar (GPR) suffers from high false-alarm rates in heterogeneous soil.

Dual-Gamma Backscatter Principle:

- Employs Compton backscattering and photoelectric absorption contrasts.
- Sensitive to atomic number (Z_{eff}) and density (ρ) variations between explosive compounds and soil matrix.
- Direct, non-metallic subsurface anomaly identification.

Baseline vs. Target

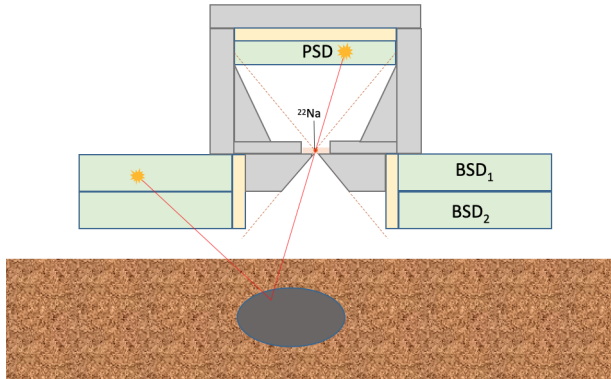
Initial Baseline:

- Conceptual 1D backscatter models.
- Coarse geometric sketches.

Current Achievement:

- Validated 3D Geant4 framework.
- Fabricated custom readout PCBs.
- Real-time FPGA processing chain.

Initial Baseline: Conceptual Multi-Detector Geometry



Initial architectural scheme: Centered ^{22}Na source, upper PSD, and dual-layer BSD arrays.

Initial Architectural Design:

- **Position Sensitive Detector (PSD):** Upper module capturing direct and high-angle backscatter events.
- **Backscatter Detectors (BSD_1 , BSD_2):** Symmetrical dual-layer side detectors to measure depth-dependent scattering.
- **Collimated ^{22}Na Injection:** Focused conical gamma ray emission toward the target depth.

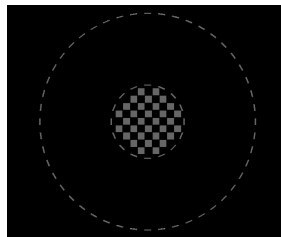
Engineering Progression

This theoretical 2D layout served as the direct basis for full 3D Geant4 modeling and discrete PCB fabrication.

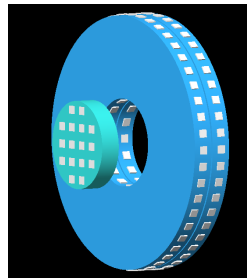
Geant4 Modeling: Scintillator & SiPM Array Architecture

Pixelated SiPM Array Topography:

- **Central Matrix:** High-density pixel grid for core photon collection and Pulse Shape Discrimination (PSD).
- **Dual Concentric Outer Rings:** Multi-channel angular discrimination for backscattered flux.
- **Crystal Interface:** Optical coupling to segmented LYSO scintillator elements for high light yield and sub-nanosecond timing.

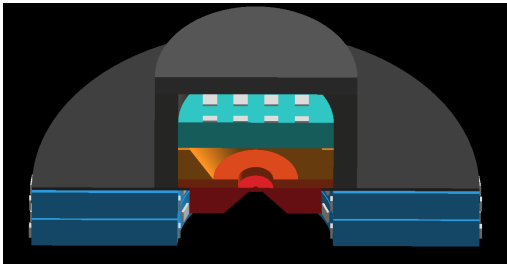


Central SiPM Matrix

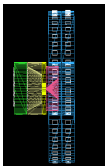


Dual Ring Coupling

Geant4 Modeling: Shielding Geometry & Collimator Assembly



Layered cutaway: Tungsten/lead shielding, collimator aperture, and SiPM array.

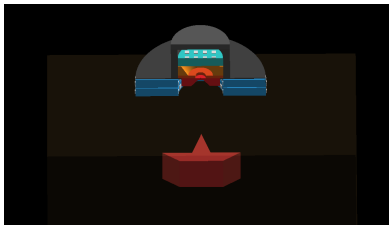


Wireframe ray collimation cone and solid angle profile.

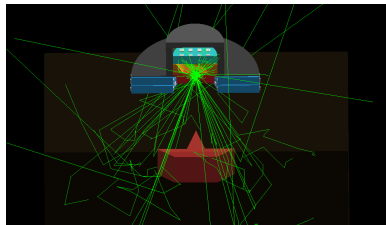
Mechanical Shielding Optimization:

- **Collimator Angle:** Optimized conical profile to project gamma rays directly onto subterranean volume.
- **Direct Beam Isolation:** High-Z (Pb/W) central core blocks primary emitter photons from hitting detectors directly.
- **Optical Isolation:** Dedicated shielding channels eliminate internal optical/radiation cross-talk.

Geant4 Ray Tracing: Subsurface Scattering Dynamics



Target setup: Star-shaped AP mine buried in soil matrix under detector.



Live ray tracing: Collimated injection and subsurface Compton backscatter.

Ray Tracing Verification:

- **Complex Target:** Asymmetrical *star-shaped* AP mine modeled within a realistic soil matrix.
- **Compton Dynamics:** Primary photons penetrate soil → scatter at mine boundaries → captured by dual SiPM rings.
- **Direct Shielding:** High-Z core blocks primary flux; registered counts stem strictly from subterranean backscatter.

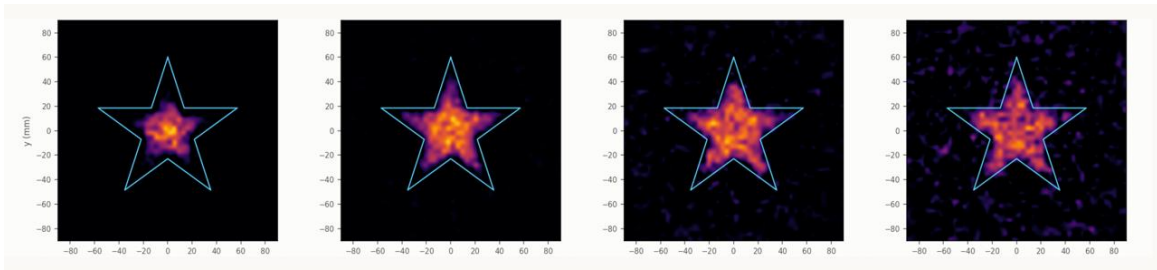
Simulation Results: 2D Spatial Reconstruction of Star Mine

Target Identification Proof:

- Progressively evaluated across increasing statistical depths and scanning passes.
- Irregular *star-shaped* AP mine geometry is cleanly resolved against surrounding soil matrix.

Physical Validation Metrics:

- Spatial contrast and energy filtering windows confirm distinct material boundaries.
- Proves detection viability for minimum-metal and complex-geometry threats.

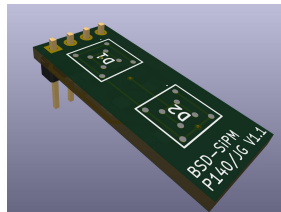


2D backscatter intensity reconstruction resolving star mine boundaries across increasing particle statistics.

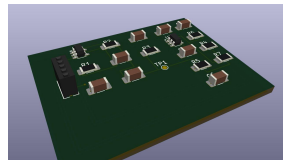
Hardware Readout: Daughter Board Architecture (BSD-SiPM V1.1)

Analog Front-End Design:

- **BSD-SiPM P140/JG V1.1:** Custom 2-channel SiPM carrier PCB for modular ring integration.
- **High-Speed Amplification:** Integrated Op-Amp conditioning stage tailored for sub-10 ns rise times.
- **Time-over-Threshold (ToT):** Active discriminator circuit generating direct digital pulses for FPGA timing.



BSD-SiPM V1.1 2-Ch Board



Active Op-Amp / ToT Stage

Hardware Fabrication: 16-SiPM Circular PSD Board & LYSO



16-SiPM PSD Board

Circular pixel matrix



LYSO Scintillator

Matched crystal block

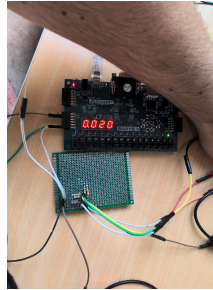
Detector Module Assembly:

- **16-Channel Array:** Circular PCB matching Geant4 pixel topography, fabricated and populated.
- **Component Population:** SMT passive filtering, decoupling arrays, and multi-pin daughter headers.
- **LYSO Scintillator:** High-density circular crystal ready for optical grease coupling.
- In-house precision hot-plate assembly and thermal verification.

Experimental Validation I: FPGA Timing & Pulse Counting

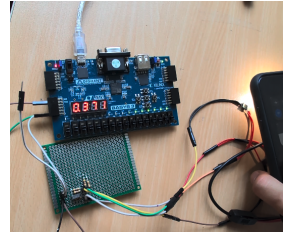
Basys 3 FPGA Interface Tests:

- Analog front-end paired with Digilent Basys 3 FPGA via high-speed GPIO headers.
- **Baseline Noise Floor:** Dark environment exhibits minimal background count rates on the 7-segment display.
- **Dynamic Triggering:** Direct optical stimulation triggers digital pulse counting logic in real time.
- Firmware discriminator and counter logic verified under live clock cycles.



Dark Test

Noise baseline count



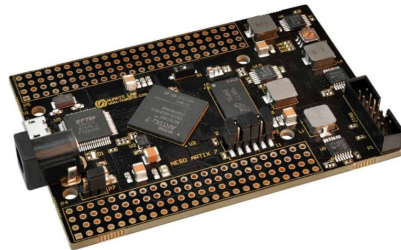
Optical Stimulation

Live pulse triggering

Hardware Readout: High-Performance FPGA Pipeline

Target Processing Core: Numato Lab Neso A7

- **FPGA Engine:** Xilinx Artix-7 FPGA (XC7A100T) delivered and actively in firmware development.
- **High-Speed Digital PSD:** Multi-channel real-time pulse shape discrimination, pulse integration, and timestamping.
- **High-Throughput Interface:** Parallel ADC/ToT interfacing to accommodate high-rate backscatter events simultaneously.
- Dedicated hardware upgrade from initial Basys 3 bench tests to final deployable core.



Numato Lab Neso A7 Board

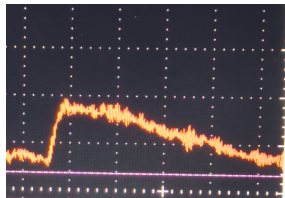
Xilinx Artix-7 (XC7A100T) dedicated readout core

Experimental Validation II: Real ^{22}Na Scintillation Signals



^{22}Na Source (740 kBq)

Shielded cavity mount



Analog Pulse Trace

Live scintillation response

Gamma Ray Response Spectroscopy:

- **Radiation Source:** Calibrated ^{22}Na gamma emitter (740 kBq) positioned within the collimator housing.
- **Oscilloscope Traces:** Direct scintillation pulses captured under dark-box conditions.
- **Pulse Dynamics:** Characteristic fast rise and exponential decay profiles validating LYSO emission physics.
- High signal-to-noise ratio established for multi-channel FPGA PSD implementation.

From Concept to Inventory: 100% Subsystem Readiness



Mass SiPM Inventory

Sensors secured for full multi-channel arrays

Principle Subsystems — All In-House:

- ✓ **^{22}Na Gamma Source:** 740 kBq calibrated emitter secured.
- ✓ **Tungsten/Lead Collimator:** Fabricated and pre-aligned.
- ✓ **Scintillator Crystals:** LYSO for PSD and BSD ready.
- ✓ **SiPM Inventory:** High-volume SiPM tray on bench.
- ✓ **Fabricated PCBs:** 16-SiPM PSD + BSD daughter PCBs.
- ✓ **Main Processing Unit:** Numato Artix-7 100T ready.

Assembly Status

Zero missing components. Transitioning directly into unified mechanical and multi-channel electrical assembly.

System Maturity Matrix & Next Milestones

Subsystem Completion Status:

Subsystem	Initial Phase	Current State
Geant4 Physics	Coarse Sketch	Validated (Star Mine)
Sensors & Crystal	Conceptual	16-SiPM PSD & LYSO Ready
Front-End PCBs	Unrouted	Fabricated & Tested
Source / Collimator	Dimensioned	In-House / Aligned
FPGA Readout	Architecture	Artix-7 100T Delivered
Firmware Pipeline	Planned	Active Development

Final 2-Month Roadmap

Month 1: Integration & Lab Testing

- Daughter PCB docking to 16-SiPM PSD array.
- Collimated ²²Na backscatter scan over soil phantom.

Month 2: Field Prototype & TRL 5

- Ergonomic casing and power unit integration.
- Final system verification and TRL 5 demonstration.